Real-Time Dynamic Path Planning via Learned First-Order Perturbations of Laplacian Potential Fields

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***Abstract*—We present a dynamic path planning framework that combines a precomputed Laplacian potential field for global guidance with a local perturbation field for online adaptation to previously unseen obstacles. Our method models obstacles and map boundaries with homogeneous Neumann boundary conditions, and updates the potential field locally whenever dynamic obstacles are detected in the robot's neighborhood. We implement our system in ROS 2 using a local costmap, a perturbation node, a waypoint planner, and a low-level controller. Three perturbation solvers were implemented: classical successive over-relaxation (SOR), a pure neural network predictor, and a hybrid method that uses a neural network to initialize SOR. A theoretical analysis shows that, under suitable assumptions, the potential field would guide the agent to travel from its initial location to its goal location without getting trapped along the path. Experimental results show that SOR is reliable but computationally expensive, the pure neural method is fast but unreliable, and the hybrid approach provides the best balance between speed and robustness. This suggests a promising novel method for path planning in dynamic environments.**

***Index Terms*—dynamic path planning, potential fields, Laplace equation, Poisson equation, robot navigation, successive over-relaxation, learning for planning**

I. Introduction

Dynamic path planning are central problems in robot planning and navigation. The robot needs to navigate to a goal in a known static map while reacting online to previously unseen moving obstacles. This setting appears in environments such as shopping malls and warehouses, where the robot knows the floor layout but must still avoid pedestrians and other dynamic agents. While classical methods such as Dijkstra's algorithm and A* are effective in static settings, they often fail in dynamic environments due to the need for expensive repeated replanning. Potential-field methods offer a promising alternative by guiding the robot through a global scalar field over free space. Perturbations to this scalar field guide the robot to avoid the previously unseen moving obstacles.

In this project, we use a Laplacian potential field for global planning in the static environment and update it locally when new obstacles are observed. Static obstacles and map boundaries are encoded through boundary conditions, while newly detected dynamic obstacles induce a local perturbation to the original field. The robot then follows the gradient of the perturbed field, allowing it to preserve global guidance toward the goal while adapting online to local changes.

We implement this framework in ROS 2 using a local costmap, a perturbation module, a waypoint planner, and a low-level controller. We compute the perturbation field using three approaches: successive over-relaxation (SOR), a neural network predictor, and a hybrid method that uses the neural network to initialize SOR. Our experiments show that SOR is reliable but slower, the pure neural method is faster but unreliable, and the hybrid approach improves speed while preserving the correctness guarantees of the classical solver. These results suggest that learning can be used to accelerate a principled PDE-based planner for dynamic robot navigation.

II. Related Work

A large body of work studies path planning as a graph-search problem over a discretized workspace. Classical methods such as Dijkstra's algorithm and A* compute shortest paths by expanding nodes in the map according to accumulated cost and heuristic guidance, respectively [3], [5]. These approaches are widely used because they are simple, complete on finite graphs, and effective in static environments. A number of extensions have been developed for replanning and real-time navigation, including Dynamic A*, D* Lite, Lifelong Planning A*, and anytime variants [8]–[10], [16].

Our work also relates to potential-field-based planning. Previous work on path planning via navigation functions has been explored extensively in the literature. Most notably, [2] first proposed the use of Laplace's equation for path planning. Extensions of this approach include solving for heat transfer functions, Hamiltonians, and artificial potential fields [1], [17], [18]. We extend this approach to dynamic path planning in this work.

III. Preliminaries

We briefly review the differential operators and boundary value problem used in this work.

Definition 1 (Laplacian)**.** *Let $\phi : \Omega \to \mathbb{R}$ be a twice continuously differentiable function on a open domain $\Omega \subset \mathbb{R}^n$. The* Laplacian *of ϕ is defined by*

$$\Delta\phi \;=\; \sum_{i=1}^{n} \frac{\partial^2 \phi}{\partial x_i^2}.$$

Definition 2 (Neumann Boundary Condition)**.** *Let $\partial\Omega$ denote the boundary of Ω, and let n be the outward unit normal*

vector on $\partial\Omega$. The Neumann boundary condition *prescribes the normal derivative of ϕ on the boundary:*

$$\frac{\partial \phi}{\partial n} = \nabla\phi \cdot n \qquad \text{on } \partial\Omega.$$

In particular, the homogeneous Neumann boundary condition is

$$\frac{\partial \phi}{\partial n} = 0 \qquad \text{on } \partial\Omega.$$

Definition 3 (Poisson Problem with Homogeneous Neumann Boundary Conditions)**.** *Let $\Omega \subset \mathbb{R}^n$ be a bounded domain with sufficiently regular boundary $\partial\Omega$, and let $\rho : \Omega \to \mathbb{R}$ be the charge density in the domain. The* Poisson problem with homogeneous Neumann boundary conditions *is to find*

$$\phi \in C^2(\Omega) \cap C^1(\overline{\Omega})$$

such that

$$\begin{cases} \Delta\phi = \rho, & \text{in } \Omega, \\ \dfrac{\partial \phi}{\partial n} = 0, & \text{on } \partial\Omega. \end{cases}$$

When a solution exists, it is unique up to an additive constant [12].

IV. Problem Formulation

Let $\Omega \subset \mathbb{R}^2$ denote the free-space workspace of the robot. Let $x_s \in \Omega$ be the robot's start location and $x_g \in \Omega$ be the goal location. The robot must navigate from x_s to x_g while avoiding both the static obstacles already encoded in the map and previously unseen dynamic obstacles that appear during execution. The objective is to generate a collision-free trajectory that reaches the goal while updating in real time as the local environment changes.

For the static environment, we define a scalar potential field $\phi : \Omega \to \mathbb{R}$ by solving the Poisson boundary value problem (Definition 3) over free space. Static obstacles and map boundaries are modeled with homogeneous Neumann boundary conditions (Definition 2), while the start and goal are modeled as source and sink charges, respectively. Following the Principle of Least Action [4], the robot then follows the gradient flow of the field

$$\dot{x}(t) = -\nabla\phi(x(t))$$

towards the goal.

When a dynamic obstacle is detected within the local region $\Omega_{\mathrm{L}} \subset \Omega$ around the robot, we solve for a local perturbation field $\Delta\phi$ over Ω_{L}, where the newly observed obstacle is imposed as an additional Neumann boundary constraint and the boundary of Ω_{L} is fixed to match the original global field. The robot then follows the gradient of the perturbed field

$$\tilde{\phi}(x) = \phi(x) + \Delta\phi(x)$$

so that it avoids the newly observed obstacle.

We make the following assumptions. First, the static environment is known *a priori* and remains fixed throughout execution. Second, the free-space map is connected, so there exists a continuous collision-free path between the start and goal in the absence of dynamic obstacles. Finally, each dynamic obstacle is fully enclosed within each local region Ω_{L}, and the dynamic obstacles do not partition Ω_{L} into disconnected components.

V. Implementation

The system is implemented in ROS 2 as a four-node pipeline consisting of a local costmap node, a perturbation node, a waypoint planner node, and a low-level controller node. The local costmap is provided by the Nav2 package [11], [14] and maintains a real-time occupancy grid of the robot's local surroundings using incoming sensor data. It identifies nearby obstacles that are not present in the original static map and passes this information to the perturbation node. The perturbation node then computes a local correction to the precomputed global potential field, the planner selects a short-horizon waypoint by following the gradient of the potential field and the controller tracks that waypoint with a proportional controller. The pseudo-code of the system is summarized in Algorithms 1, 2, 3, and 4.

Algorithm 1 Local Costmap Node (Nav2)

```
Subscribe to lidar observations
Maintain rolling local occupancy grid centered at robot
Fuse incoming sensor hits into occupied cells
Publish local costmap
```

Algorithm 2 Perturbation Node

```
Load global map and potential field φ
On local costmap update:
if all local obstacles exist in global map then
    Publish zero perturbation field Δφ = 0
    return
end if
Extract local patch φ_patch
Overlay new obstacles and inflate them
Mark obstacles and boundaries as fixed
Solve Laplace on patch:
if solver = SOR then
    Run SOR iterations on φ_patch
    Δφ ← φ_patch − φ_global
else if solver = NN + SOR then
    Predict Δφ with NN, refine with SOR
else if solver = NN only then
    Predict Δφ with NN
end if
Publish Δφ
```

Algorithm 3 Waypoint Planner Node

Load potential field ϕ
Initialize $\Delta\phi \leftarrow 0$
Set goal threshold ϕ_{goal}

On each timer tick:
$(r, c) \leftarrow$ current grid cell
if $\phi[r, c] \geq \phi_{\text{goal}}$ **then**
    return
end if
for $i = 1$ to lookahead **do**
    Compute local gradient $\mathbf{g} \leftarrow -\nabla(\phi + \Delta\phi)(r, c)$
    $(r, c) \leftarrow (r, c) + \eta\,\mathbf{g}$
end for
Publish waypoint (r, c)

On perturbation update:
    Update $\Delta\phi$

Algorithm 4 Robot Controller Node

Set gains $k_{\text{lin}}, k_{\text{ang}}$ and tolerance ϵ

On waypoint update:
    Set goal (x_g, y_g), activate goal

On odometry update:
if no active goal **then**
    Publish zero velocity
    return
end if
Compute distance $d \leftarrow \|(x, y) - (x_g, y_g)\|$
if $d < \epsilon$ **then**
    Clear goal, publish zero velocity
    return
end if
Compute heading error θ_e
$\omega \leftarrow k_{\text{ang}} \cdot \theta_e$
$v \leftarrow k_{\text{lin}} \cdot d \cdot \max(0, \cos(\theta_e))$
Publish (v, ω)

Three solvers were implemented for solving the local perturbation patch. In the classical **SOR** mode, the local patch is solved directly with successive over-relaxation (SOR) [19] after inserting newly observed obstacles into the patch. It explicitly enforces the boundary conditions of the local patch and ensures the local potential field's continuity with the surrounding global field. However, in dense environments, convergence can require many iterations, thereby resulting in high computational costs.

The learning-based modes attempt to reduce this cost by predicting the perturbation field with a neural network rather than computing it using SOR. In the **pure NN** mode, a ResNet-style convolutional network [7] predicts the perturbation field directly. The network architecture is depicted in Figure 1. In practice, this approach was unreliable and often caused the robot to collide with the dynamic obstacle. To address this, the **NN Init + SOR Refinement** mode uses the network only

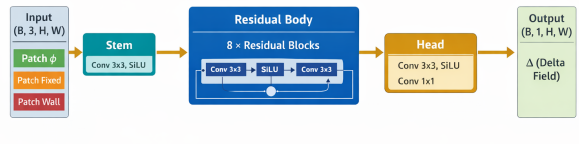

Fig. 1. Architecture for **pure NN** mode[1]

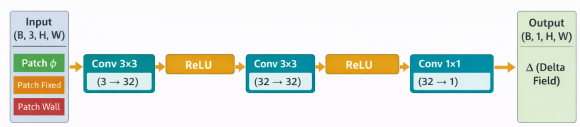

Fig. 2. Architecture for **NN Init + SOR Refinement** mode[2]

to produce an initial guess, which is then refined by SOR until convergence. The network is a simple Multilayer Perceptron [6] for fast inference and is depicted in Figure 2.

VI. Theoretical Analysis

This section provides an analysis of the theoretical properties of our proposed methodology. The main result is that, under certain assumptions, if a collision-free source-to-sink path exists, then an agent following the perturbed potential field will continue progressing from the source to the sink without becoming trapped. The key idea is that the global potential field induces source-to-sink field lines, while a bounded local harmonic patch cannot create interior terminal points. This is formalized more rigorously in the following lemmas and theorem.

Lemma 1 (Field Lines Connect Source to Sink)**.** *Let $\Omega \subset \mathbb{R}^2$ be a bounded, connected open set with boundary $\partial\Omega$, and let ϕ satisfy*

$$\nabla^2\phi = -\rho \qquad \text{in } \Omega,$$

with Neumann boundary condition

$$\nabla\phi \cdot n = 0 \qquad \text{on } \partial\Omega.$$

Assume $\rho \in L^1(\Omega)$ satisfies

$$\rho > 0 \quad \text{on } \Omega_s,$$
$$\rho < 0 \quad \text{on } \Omega_t,$$
$$\int_\Omega \rho\, dx = 0,$$

where $\Omega_s, \Omega_t \subset \Omega$ denote the source and sink regions. Define

$$\mathbf{E} = -\nabla\phi.$$

Then the integral curves (field lines) of $\mathbf{E}$:

- *originate at source regions,*
- *terminate at sink regions,*
- *and do not form closed loops or terminate in free space.*

[1]Image generated using AI.
[2]Image generated using AI.

Proof Sketch. We rule out all other possible behaviors of the field lines.

1) **Field lines cannot stop in free space.** If a field line terminated at a point where $\rho = 0$, then this would imply a local extrema of ϕ in a region where $\nabla^2\phi = 0$, which contradicts the maximum principle of harmonic functions.
2) **Field lines cannot form closed loops.** Since $\mathbf{E} = -\nabla\phi$ is a gradient field, it is curl-free:
$$\nabla \times \mathbf{E} = 0.$$
Hence line integrals are path-independent, so closed field-line loops cannot occur.
3) **Field lines can only start or end at sources and sinks.** By Gauss's law,
$$\nabla \cdot \mathbf{E} = \rho,$$
so divergence is nonzero only in source and sink regions. Therefore, field lines may only originate where $\rho > 0$ and terminate where $\rho < 0$.
4) **Field lines do not cross the boundary.** The Neumann condition $\nabla\phi \cdot n = 0$ implies that $\mathbf{E}$ is tangent to $\partial\Omega$, so field lines neither enter nor leave through the boundary.

Therefore, the only remaining possibility is that field lines connect source regions to sink regions. □

Lemma 1 establishes the global source-to-sink structure of the field.

Lemma 2 (Boundary-to-boundary field lines in a bounded source-free region)**.** *Let $\Omega_L \subset \Omega$ be a bounded, connected, source-free region, and let ϕ satisfy*
$$\nabla^2\phi = 0 \qquad \text{in } \Omega_L,$$
with boundary conditions on $\partial\Omega_L$ fixed by the surrounding solved global potential. Define
$$\mathbf{F} = -\nabla\phi.$$
If ϕ is nonconstant in Ω_L and has no interior critical points, then every field line of $\mathbf{F}$ in Ω_L connect boundary points of $\partial\Omega_L$.

Proof Sketch. Since ϕ is harmonic in Ω_L, the maximum principle implies that ϕ has no interior maxima or minima unless it is constant. Thus the flow cannot be trapped by an interior extremum. Since ϕ has no interior critical points, the vector field
$$\mathbf{F} = -\nabla\phi$$
is nonvanishing in the interior of Ω_L, so field lines cannot start or end there. Because Ω_L is bounded and connected, the only remaining possibility is that each field line enters and exits through the boundary. Hence field lines in Ω_L connect boundary points of $\partial\Omega_L$. □

Lemma 2 shows that a bounded local harmonic patch cannot create interior terminal points that trap the agent in its interior.

Theorem 1 (No trapping under harmonic field following)**.** *Let $\Omega \subset \mathbb{R}^n$ be a connected domain, and let*
$$\phi \in C^2(\Omega \setminus (\Omega_s \cup \Omega_t)) \cap C^0(\Omega)$$
solve
$$\nabla^2\phi = -\rho \qquad \text{in } \Omega,$$
with homogeneous Neumann boundary conditions
$$\nabla\phi \cdot n = 0 \qquad \text{on } \partial\Omega,$$
where $\rho \in L^1(\Omega)$ satisfies
$$\rho > 0 \quad \text{on } \Omega_s,$$
$$\rho < 0 \quad \text{on } \Omega_t,$$
$$\int_\Omega \rho \, dx = 0,$$
and $\Omega_s, \Omega_t \subset \Omega$ denote the source and sink regions, respectively. Define the flow field
$$\mathbf{F} = -\nabla\phi.$$

Let $\Omega_L \subset \Omega$ be a bounded, connected, source-free subregion. Assume that any dynamic obstacles contained in Ω_L are fully enclosed, do not partition Ω_L, and that the induced local harmonic field in Ω_L has no interior critical points.

Then an agent following the field lines of $\mathbf{F}$ moves from the source region to the sink region without becoming trapped inside Ω_L.

Proof Sketch. Let a be any point on the field line followed by the agent.

Case 1: $a \notin \Omega_L$. Then a lies in the global free-space region. By Lemma 1, global field lines connect source regions to sink regions and do not terminate in free space. Hence the agent continues along the same source-to-sink field line.

Case 2: $a \in \Omega_L$. Then a lies in a bounded, connected, source-free region. By Lemma 2, field lines cannot originate or terminate in the interior of Ω_L, and must intersect the boundary $\partial\Omega_L$. Thus, the agent cannot get stuck inside Ω_L and must exit the region and rejoin the global source-to-sink flow outside Ω_L.

Since this argument holds for every point a along the trajectory, the agent continues along a collision-free source-to-sink field line and does not get stuck. □

The theorem therefore guarantees that under ideal conditions, the agent would travel from its initial location to its goal location without getting trapped along the path. In practice however, discretization and partial observability may violate these assumptions. Such shortcomings would be further addressed in the limitations section.

Fig. 3. `Bookstore_World` model

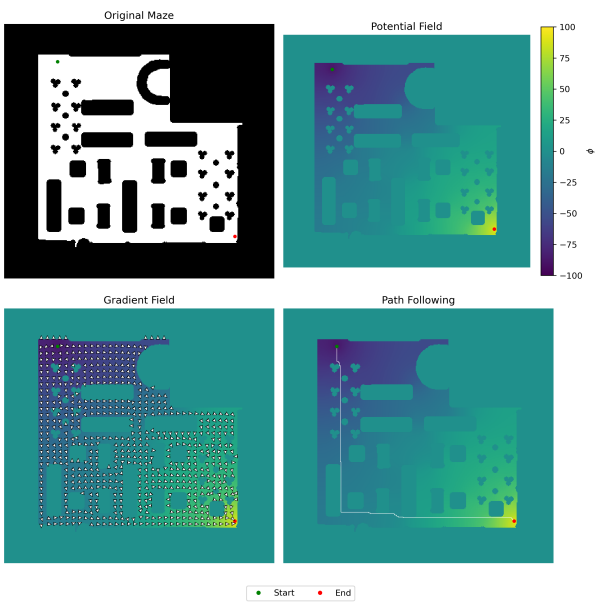

Fig. 4. Potential field, gradient field, and path taken in static map

VII. Experimental Evaluation

We evaluate our dynamic path planning framework in simulation. The robot is modeled as a TurtleBot Burger with a 360° field of view and a sensing radius of 2 meters. The environment is constructed by extending AWS Robomaker's `Bookstore_World` model in Gazebo [15]. Three human obstacles are inserted at locations that are not represented in the original static map (Figure 3). The global potential field is precomputed from the static environment (Figure 4), while the local costmap detects nearby disturbances online and triggers the perturbation solver when previously unseen obstacles enter the robot's neighborhood.

For the classical **SOR** solver, the perturbation field is computed by explicitly solving the local Laplace patch after inserting the newly observed obstacles. As expected, the robot robot detected the new obstacles and adjusted its global path accordingly. However, its primary drawback is computational cost. Each call to the **SOR** solver took an average runtime of 215 ms and 434 iterations of over-relaxation. In dense environments, this often exceeded 800 ms and over 2000 iterations.

Motivated by this drawback, the **pure NN** solver predicts the perturbation field directly using a learned convolutional network rather than having it solved numerically. Although its inference speed is fast, it was unsuccessful in practice. The predicted perturbation fields were not sufficiently accurate, and the robot consistently collided with the dynamic obstacle in the simulation trials. This suggests that a purely learned perturbation lacks the theoretical guarantees needed for safe online navigation, and depends heavily on the chosen architecture and prediction quality.

To address this tradeoff, the **NN Init + SOR Refinement** solver uses a neural network to predict an initial candidate perturbation field, which is then refined using SOR until convergence. This hybrid approach preserves the correctness and convergence behavior of the classical numerical solver while reducing its computational cost. In our experiments, each call to the hybrid solver required an average runtime of 164 ms and 317 iterations, corresponding to speedups of approximately 24% and 27%, respectively, relative to SOR. Thus, the hybrid approach accelerates convergence rather than replace the iterative solver outright, and retains the theoretical properties of the classical approach as a lower bound on performance. The visualizations for our experiments can be found here.

VIII. Limitations

Three key limitations were identified for our proposed method. First, the theoretical analysis is posed in a continuous and analytic domain, whereas the implemented planner operates on a discretized occupancy grid. This means that the solved field is only an approximation of the ideal harmonic potential, and could result in the robot getting stuck. Second, dynamic obstacles are only partially observable from onboard sensing. For example, the lidar cannot observe the "back side" of an obstacle. This means that the obstacle may not be fully enclosed within the local perturbation region, which can cause field lines to route into unseen occupied space. Finally, neither the robot nor the obstacles are point masses, and obstacle inflation can effectively partition the local neighborhood, thereby violating the connectivity assumptions required by the theory.

IX. Future Work

Several directions remain for future work. First, the current method could be extended with better obstacle modeling, such as predicting the future motion of dynamic obstacles from lidar observations rather than reacting only to their current detected shape. This would allow the perturbation field to account for anticipated obstacle motion and potentially improve safety in crowded environments. Second, the pure learning approach

could be improved by exploring alternative neural architectures and training strategies, since the current network was not sufficiently accurate for reliable standalone use. Finally, an interesting direction would be to explore physics-informed neural networks (PINNs) for perturbation prediction [13]. This approach could potentially improve the physical consistency of our prediction while still retaining the speed advantages of zero-shot inference.

X. Conclusion

In this work, we presented a dynamic path planning framework that combines a precomputed static Laplacian potential field for global guidance with a local perturbation field for online adaptation to previously unseen obstacles. Under idealized conditions, the framework provably guarantees that the agent would travel from its initial location to its goal location without getting trapped along the path. Empirically, a classical SOR-based approach is reliable but computationally expensive, a purely learned perturbation is fast but unreliable, and a hybrid learned initialization with SOR refinement provides the best balance between speed and robustness. Overall, these results suggest that using learning for speeding up PDE-based potential field guidance is a promising method for dynamic path planning as compared to conventional graph traversal path-finding algorithms.

Contribution Statement

This project is completed as part of the Final Course Project for CMU 16-832 Integrated Planning and Learning (Spring 2026). The problem formulation, algorithm design, implementation, experiments, and writing were carried out by Boxiang (William) Fu. The code repository for this project is available at https://github.com/williamfbx/Laplace-Pathfinder. The experiment visualizations for this project is available in our Google Drive.

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